Manifold Gauge



Category: **Measuring & Marking Tools** Type: **Full SOP** Priority: **High** 3 labs

Covers these lab items: Manifold Gauge; manifold gauge sets

What it is used for

A manifold gauge set measures pressure in compatible fluid systems and can support diagnosis, evacuation, and charging when used with the correct service procedure. Correct gauge selection, hose connections, pressure limits, and refrigerant-specific handling are essential to prevent leaks, equipment damage, and injury.

Main parts

- Low-pressure gauge displays pressure on the low side of the system.
- High-pressure gauge displays pressure on the high side.
- Manifold body connects gauges, valves, and service ports.
- Control valves regulate flow through the manifold.
- Service hoses connect the manifold to approved system ports or service equipment.
- Couplers or adapters provide compatible connections.
- Sight glass shows fluid movement where fitted.
- Protective caps help keep service ports clean.

Before you start

1. Only trained personnel should use the manifold gauge set for the relevant system.
2. Identify the system type, working fluid or refrigerant, and manufacturer service specifications.
3. Confirm the gauges, hoses, seals, and couplers are rated for the expected pressure and compatible with the fluid.
4. Inspect gauge faces, hoses, valve stems, fittings, and seals for damage, wear, or leakage.
5. Use required PPE and ensure adequate ventilation, especially when working with refrigerants or other pressurised fluids.
6. Confirm that the manifold valves are in the required initial position and service ports are accessible.
7. Use only the approved refrigerant recovery and charging equipment where applicable; do not vent refrigerant intentionally.

How to use

1. Switch off or prepare the system according to the manufacturer's service procedure.
2. Identify the correct low- and high-pressure service ports before connecting hoses.
3. Connect the hoses using compatible couplers and the prescribed connection sequence.
4. Check that connections are secure and avoid cross-connecting the high and low sides.

5. Read the static or operating pressures only under the conditions specified by the service procedure.
6. Operate manifold valves only when the procedure requires it and understand which flow path each valve controls.
7. Monitor pressure changes and compare readings with the system specifications and operating conditions.
8. Stop the procedure if a hose leaks, a gauge behaves abnormally, or pressure approaches the equipment limit.
9. Recover or transfer fluid using approved equipment and the specified procedure where required.
10. Close valves, isolate the system as instructed, and disconnect hoses carefully to minimise fluid release.
11. Refit service-port caps, inspect for leaks using an approved method, and record the readings and work performed.

⚠ Safety

- Never exceed the rated pressure of any gauge, hose, coupler, or adapter.
- Do not connect a gauge set to an unknown or incompatible fluid system.
- Do not loosen pressurised connections or disconnect hoses before following the required isolation and recovery procedure.
- Never release refrigerant deliberately into the atmosphere; follow applicable recovery and environmental rules.
- Keep the set away from ignition sources when working with flammable refrigerants.
- Wear suitable eye and hand protection and avoid contact with escaping cold fluid or vapour.

Care & storage

- Depressurise and store the set according to the manufacturer's instructions after use.
- Keep hoses capped and protect gauge faces, fittings, and couplers from impact and contamination.
- Inspect seals, hoses, and valve operation regularly and replace worn components with compatible parts.
- Arrange calibration when required or when readings are inconsistent with a reliable reference.
- Store the set in a clean, dry case away from excessive heat and corrosive chemicals.

Common mistakes to avoid

- Using gauges or hoses with an inadequate pressure rating or incompatible fluid rating.
- Connecting the high-pressure hose to the wrong service port.
- Opening manifold valves without understanding the system flow path.
- Disconnecting hoses without the prescribed isolation or recovery steps.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.